Philosophical Logic Essay Notes

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Contents

1 Metaphysical Necessity

BROAD THEME: What is the correct logic for metaphysical necessity?

Core Reading
Alvin Plantinga, The Nature of Necessity (OUP, 1978), Chapter I
Conceptions of necessity?

- There are true propositions but some are contingent while some are necessary
  - "the average annual rainfall in Los Angeles is about 12 inches" is contingent
  - "7+5=12" or an application of modus ponens are necessary
- How do we distinguish between necessity and contingency?
  - Not meaningful/informative to say that p is necessary iff its denial is impossible
  - *Logical necessity*: logical necessity are the truths of propositional logic or FOL
  - Platinga suggests for a *wider conception of necessity* i.e. "broadly logical necessary" like the truths of set theory, arithmetic and mathematics or propositions like "Red is a colour" or "No one is taller than himself" or "No numbers are human beings"

    Why? We want such a broad conception because it makes sense of philosophical debates over the status of propositions such as "Every person is conscious at some time or other"
  - Platinga's wider notion of necessity is narrower than *causal or natural* necessity expressed by propositions like "Voltaire once swam the Atlantic" which is naturally impossible
  - Necessity vs "unrevisability": some philosophers like Quine argue that no proposition is immune from revision (in particular, Quine thinks that this includes the law of non-contradiction or modus ponens should they be empirically contradicted).

For Quine says: "it becomes folly to seek a boundary between synthetic statements which hold continently on experience, and analytic statements, which hold come what may...no statement is immune to revision...the logical law of excluded middle has been proposed as a means of simplifying quantum mechanics; and what difference is there between such a shift, and the shift whereby Kepler superseded Ptolemy"

But even if we concede that every statement is subject to revision, it does not follow that there are no genuine necessary propositions.

Why? Platinga thinks that such philosophers confuse necessity as a trait endowed via popular favour; even if people hold a principle to be false, a proposition could still be necessarily true.

Reply: I do not think that Platinga is quite right here – surely what Quine should be taken to mean is that our empirical experiences can render "analytic" statements false and therefore imply that they are, in fact, contingent
Platinga takes his conclusions further to suggest that necessity must be distinguished from what cannot be *rationally rejected* i.e. ZFC might be necessarily true even if it is rational for me to give it up should I hear rumours of its failure.

Remark: Makes sense since rationality makes no commitment on success.

- Necessity vs the self-evident and a priori: the latter two are epistemological concepts, and are about how we come to know propositions.

  A self-evident proposition p is a proposition that is utterly obvious to anyone who understands it e.g. modus ponens. But Platinga thinks that "97+342+781=1220" is necessary but not self-evident. Is Platinga taking the notion of self-evident-ness too literally?
  Platinga thinks that the border between the self-evident and necessary is clear even under an extended understanding: that it is a consequence of self-evident truths by argument forms whose corresponding conditions are themselves self-evident i.e. can be deducible from self-evident truths.
  Counter-example: the Continuum Hypothesis is either necessarily true or necessarily false but there is little reason to think that its acceptance or denial are deducible from self-evident propositions by self-evident steps. So there are propositions that are necessarily true but not self-evident.
- Necessity vs the a priori (cleavage between a priori and a posteriori): the latter is a distinction in epistemological terms while the former distinction between necessity and contingency.

  Parallel between a priori and self-evident: Fermat's Last Theorem or its denial is a necessary truth but are not known at all, and therefore not known a priori.

  Is every necessary truth that is known known a priori?
  (a) Is every necessary truth that is known, known a priori to everyone who knows it? Platinga suggests that this is not true, since I can come to know some theorem a priori by learning the proof, but also a posteriori via learning it from testimony
  (b) Is every necessary truth that is known to someone or other, known a priori to someone or other? Platinga suggests that this is not so clear, but argues that there are example propositions that are necessary but probably not known a priori to any of us.

- Modality De Dicto and Modality De Re

  An assertion of modality de dicto, for example "necessarily nine is composite", predicates a modal property – in this case necessary truth – of another dictum or proposition, in this case "nine is composite".

An assertion of modality de re ascribes the *necessary or essential possession of the property* i.e. of *being composite*.

Example: Suppose we have two statements (a) Every human being is necessarily rational and (b) Every animal in this room is a human being. Modality de re sanctions the inference (c) Every animal in this room is necessarily rational, while rejecting the inference that (d) Every human being is necessarily in Australia from (e) Every rational creature is in Australia and (f) Every human being is necessarily a rational creature. Furthermore, (c) is not to be read as "It is necessarily true that every animal in this room is rational"

The difference: The ascription of some necessary or essential property (being rational) does not license us to suggest the essentiality of another property (being in Australia). Furthermore, having a certain property necessarily cannot be construed as having the property being a necessary truth. In short, the expression of something having a certain property has necessary (perhaps to its identity as that object) is an expression of modality de re rather than modality de dicto.

What is, in fact, modality de re? Presumably, it means that the object in question could not conceivably have lacked the property in question; that under no possible circumstances could the object have failed to possess that property.

Reply: Then, why is it not necessarily true that the object would have the property in question if it obtains in every possible world/circumstance?

Platinga suggests that no definition of modality de dicto or de re is to be of use – need example instead. While 5 being prime is an essential property, and suppose I am thinking of the number 5. The proposition "What I am thinking of is prime" is not necessary true but "What I am thinking of is prime" is essential to what I am thinking of.

Aquinas suggests that a given statement of modality de dicto may be true when the corresponding statement of modality de re is false. For instance "What I am thinking of is essentially prime" is true but "Necessarily, what I am thinking of is prime" is false.

Some philosophers think that the idea of morality de dicto is tolerably clear but argue that modality de re is a source of boundless confusion.

Nathan Salmon, The Logic of What Might have Been, The Philosophical Review 98 (1989), 3–34

- Main Thesis: S5 and S4 (and also B) emobdy an invalid pattern of modal reasoning, and instead that T is the one and only (strongest) correct system of modal logic.

  Attacks the S4 Axiom Schema: $\Box\phi \rightarrow \Box\Box\phi$ or that for any necessarily true proposition p, it is necessarily true that itself is necessarily true. A foritori, S5 fails since the S4 axiom is provable in S5. In terms of accessibility relations, we want to say that $\mathscr{R}$ is NOT transitive (nor symmetric).

  For Salmon, the only appropriate constraint on $\mathscr{R}$ is reflexivity.

  Salmon's motivating example is that, for some necessary truths p, like that a certain table does not originate from a certain chunk of wood, the fact that p is necessary cannot itself be correctly deemed necessary. Instead, the claim that a necessary truth p is in fact necessarily true is false. Accordingly, $\Box\Box\ldots\Box p$ is false, even if $\Box p$ is true.

  Also rejects the system B, characterised by the axiom schema $\phi \rightarrow \Box\Diamond\phi$ i.e. for any true proposition, the proposition that p is possibly true is itself necesarily true. This is because that, even if the B axioms are in fact true, and even if they are necessarily true, it seems to be logically possible for some proposition p to be true while the proposition that p is necessarily possible at the same time false. So, the B axiom's logical truth remains in need of justification.

- Case Against S4: The main intuition is that a particular wooden table, called "Woody" could have originated from matter slightly different from its actual original matter m^* but not entirely different matter.

  We may stretch things to the limit by selecting some matter m such that, although Woody could not have originated from m, m is close enough to being possibility for Woody that if Woody had originated from certain matter m' that is in fact possible for Woody i.e. it is possible for Woody to have originated from m' relative to its actual matter m^* and possible for Woody to have originated from m relative to its possible matter m', but it is NOT possible for Woody to have originated from m relative to m^*. In short, the relation that Woody possibly originates from an arbitrary material m_2 from m_1 is not tranitive.

  Salmon argues that possible words are both (i) the standard identification of necessity with truth in every possible world and possibility with truth in at least one possible world and (ii) a certain sort of maximal abstract entities according to which certain things (facts, states of affairs) obtain and certain other things do not obtain.

  - Lewis: Possible worlds are total ways things might have been
  - Kripke: A possible world is something like a total history that might have obtained concerning everything in the cosmos
  - Kripke, Stalnaker: A maximal property or state that the cosmos might have had or been in
  - Salmon: Maximal Scenario that might have obtained i.e. an infinitely long, complex and detailed set of states of affairs or (potential) facts or statements

  Essentially, a possible world does not leave any comprehensive range of questions of fact undecided.

  $\mathscr{R}$ is not transitive from example: Woody cannot be in the state of originating from m i.e. originating from m is metaphysically unavailable to Woody – it is a way that Woody cannot be. But it is still a way for an individual to be. Let us call this maximal way for things to be "W", and since Woody originates from m according to W and Woody metaphysically cannot do so, W is a total way things cannot be. Since total ways things cannot be is a total way for things to be such that things cannot be that way, a state or history for everything in the universe s.t. everything in the universe cannot be in that state or have that history, a maximal state of affairs or scenario that cannot obtain. Thus, total ways things cannot be are also "worlds". So, there are impossible worlds (relative to other worlds).

  But although W is an impossible world, there is a possible world W' according to which Woody originates from m' instead of m^* and if W' had obtained, W would have been a way things might have been rather than a way things cannot be; W would thus be possible instead of impossible.

  Key thesis: Although W is impossible relative to the actual world, it is possible relative W', which is itself possible relative to the actual world. Thus W is a possibly possible world. Other impossible worlds may be not even possibly possible world. Other impossible worlds may be not even possibly possible, but not possibly possibly possible, and so on. The possibly possible is not equivalent to possibility simpliciter (S4$\Diamond$ axiom). So, the binary relation between worlds of relative possibility – accessibility – is not transitive.

  Reply: Is Salmon not begging the question with his intuitions? He thinks that the possibility of origination is not transitive and uses it to show that possibility itself is not transitive. There is no positive reason to support this.

- Objection I (Lewis): "This is no defence [of the essentialist doctrine that a table could not have originated from entirely different matter], this is capitulation [to radical anti-essentialism]. In these questions of haecceitism and essence, by what right do we ignore worlds that are deemed inaccessible? Accessible or not, they're still worlds. We still believe in them. Why don't they could?

  Salmon: Intransitive accessibility relations are introduced into modal semantics for the purpose of interpreting various "real" or restricted types of modalities, such as nomological necessity.

  – Nomological necessity: A proposition is *nomologically necessary* in an arbitrary possible world w iff it is true in every possible world in which all of the laws of nature in w are true.
    We may say that a world w' is accessible to, or nomologically possible relative to, a world w if every natural law of w is true in w'.
    More succinctly: a proposition is physically necessary with respect to a possible world w iff it is true in every possible world accessible in which all of the law of physics in w are true. More restricted modalities require such alternative accessibility relations as suggested by Salmon

  But the hallmark of metaphysical necessity or necessity tout court is **a complete lack of restriction for what is accessible to us**. Any possible world is possible in the unrestricted, metaphysical sense.

  Lewis uses this to defend S5 modal logic, which is characterised by an equivalent relation as $\mathscr{R}$.

- Salmon's Reply to Lewis: Lewis conflates two distinct notions: (a) the *generic notion* of a way for things to be and (b) the peculiarly *modal notion* of a way things *might have been*.

  The Generic Notion: underwritten by the layman's understanding of a possible world almost in terms of a planet or a universe.

  The Modal Notion: the metaphysics of modality describes a world as an abstract entity according to which some things obtain and other things do not. They are not *physical universes* but *intensional entities* that represent things as being one way or another.

  Remark: Does this distinction actually hold water? Can the layman understanding actually differ from the modal notion i.e. could we say that the layman understanding is a heuristic that does not improperly describe what possible worlds are?

  How this confusion undermines Lewis's point: Lewis maintains the layman, generic conception of a world as a whole physical universe and combines this conception with the metaphysician's conception of a world as an entity according to which some states of affairs obtain.

  This combination makes us think that a possible world is a way for things to be that might have existed. But Salmon suggests that it is not, and instead a way for things to be such that things might have been that way. I.e. a possible history for an individual is not a history that might have existed, but a history that the individual might had or might have been in. In turn, the word "possible" contribute some special meaning to the phrase "possible world", more so than is accommodated by the generic notion of a total way-for-things-to-be-even-if-things-could-not-have-been that way.

  In short, Salmon thinks that the modal (metaphysical) notion of possible words is more restricted than the generic notion i.e. a modally special kind of total way for things to be (a way that things *might have been*). So, not mere possible existence, and metaphysical modality is NOT the limiting case of restricted modalities or that metaphysical necessity and possibility is unrestricted necessity or possibility.

  More modalities in Plato's heaven than are dreamt of in one's philosophy. Many metaphysical impossibilities are mathematically possible, for example, Nathan Salmon being a Visa credit card account with the Bank of America.

Salmon thinks that logical necessity (true on logical grounds alone) is less restrictive than metaphysical modality. While it might be logically necessary that Nathan Salmon is not somebody other than Nathan Salmon, it is also logically necessary that either Nathan Salmon is a Visa credit card account with the Bank of America or he is not, it is not logically necessary that Nathan Salmon is not a credit card account (which is metaphysically necessary according to Salmon).

Remark: I don't think this makes much intuitive sense.

In what sense is metaphysical necessity restricted? Salmon pedantically says that metaphysical necessity is restricted to world that are metaphysically possible.

Tim Williamson, Identity and discrimination, reissued and updated edition (Wiley- Blackwell, 2013), Chapter 8
Basic Motivation: A criticism of Salmon-style examples against the transitivity of possibility because they do not generalise to all modal and temporal paradoxes – in fact, the underlying structure of the counterexamples is similar to one of personal identity across time.

Williamson's modal paradox

- Salmon style 'Sorities' paradoxes: begin with an artifact x originally made from material m_0 in the current world w_0 and that x could have been originally made of slightly different material, so there is a possible world w_1 in which x is originally made of m_1, 99% of which is m_0. By parity of reasoning, there is such a world w_2 in which x is originally made of m_2, 99% of m_1, but only 98% of which is m_0. We can continue to w_{100} where x is made of m_{100}, none of which is part of m_0, so x could not have been made out of an utterly different material
- Solution from Salmon is to distinguish "possibly possible" from merely "possible" i.e. w_3 is possible from a possible world w_2 relative to w_1 but w_3 is not possible from w_1.

  Basically, Salmon accepts the idea of diminishing possibility from some fixed world across possible worlds
- Earring paradox: Basically, suppose an earring maker makes an earring by cutting a metal circle across some diameter d. Let us denote the point at which one cuts at the circle $o(r)$ for some r where the earring you get from cutting at $o(r)$ is the upper half (call this the left earring). Now, by the Sorities paradox, we have $o(0) = o(180)$ but clearly $\sim o(0) = o(180)$ since otherwise, you can wear the exact same earring on both your left and right ears.

Basic solution: Our reference of 'that earring' or 'that Rome' is indeterminate, not the identity relation

- Identity is an equivalence relation and hence transitive; it is also a determinate relation
- If it is the case that identity is transitive or = is transitive, and $\sim o(0) = o(180)$ after the Sorities chain, it is maybe that for some r, $o(r) \neq o(r+1)$ i.e. the chain breaks down somewhere

  Williamson points out that holding this would be completely arbitrary
- We cannot easily discriminate between when the earring ends and when it begins. Williamson thinks that we are not speaking of the identity relation during the Sorities paradox but what he calls an M paradox

  For instance, $o(x) = o(y)$ iff $|x - y| \leq 90$ for the earring case. Clearly, M is reflexive and symmetric but NOT transitive

- We cannot identify ear-ring identity with M because M is only a necessary but insufficient condition for earring identity, but we can try to save the original conception via maximal M relations
- Williamson suggests that we can have a maximal M-relation that induces a partition into equivalence classes i.e. $\{[0, 90), [90, 180), [180, 270), [270, 360)\}$; but this is problematic since we can add 5 to each
- It is basically implausible to find a maximal M-relation to resolve our conundrum and our paradox
- Solution: We can never know at which juncture $(j, j+1)$ does the identity fail and because of that there is a relevant indeterminancy of reference – the paradox arises from our assumption that at every interval $(j, j+1)$, 'that earring' refers to the same earring when unbeknownst to us, passed a certain point, the referent changes
- Similar to temporal paradoxes about when old Rome ends and new Rome begins – we still refer the different objects under the same name of 'Rome' and assume it refers to the same thing at every small interval of time $(t, t+\varepsilon)$ when for some t, it does not.

Tim Williamson, Modal science, Canadian journal of philosophy 46, 453-492. Sections 1 and 2.
Basic argument: metaphysical necessity is the strongest 'objective necessity'

Objective modalities

- 'objective modalities are envisaged as out there in the world, independently of us'
- 'objective modalities are non-epistemic, non-psychological, non-intentional'

> Let 'n' name the actual number of inhabited planets... even though we know for sure that there are no fewer than n inhabited planets, there could have been fewer than n, because there could have been no inhabited planets. Such a sense in which things could have been otherwise is objective rather than epistemic.

There is basically an intentional difference between 'the number of planets' and 8. Following Kripke on the standard metre, the phrase 8 is a rigid designator across *all possible worlds*, while it just so happens that at world w_1, 8 is the number of planets.

- Objective necessities (ON) come in many varieties
- MN is the strongest ON:
  - p is metaphysically possible iff 'it has at least one sort of objective possibility'
  - p is metaphysically necessary iff 'it has every sort of objective necessity'

Williamson's argument is in two segments:

- Technical part: the logic of MN is S5
- Philosophical part: rebuts scepticism about MN

Logic of MN is S5: Given some assumptions about (i) propositions and (ii) ONs, Williamson proves that the logic of MN is (at least) S5 **Assumption (i): a propositional Booleanism**

- Williamson assumes (a) the propositions make up a Boolean algebra, and (ii) any set of propositions has a conjunction
- Truth values: The truth values make up a Boolean algebra $\{0, 1\}$ with $\{\wedge, \vee, \sim\}$ being expressively adequate truth functions

- Sets of worlds: the subsets of W make up a complete Boolean algebra i.e. $\{\wedge, \vee, \sim\}$ are $\{\cap, \cup\}$ and (relative) complement i.e. $W \neq \emptyset$. For $p, q \subseteq W$:

$$\sim p = W \backslash p = \{w \in W : w \notin p\}$$
$$p \wedge q = p \cap q = \{w : w \in p \text{ and } w \in q\}$$
$$p \vee q = p \cup q = \{w : w \in p \text{ or } w \in q\}$$

  The tautology, $1 = W$; the contradiction, $0 = \emptyset$
  p entails q iff $p \subseteq q$ iff $p \cap q = p$ iff $p \cup q = q$.

- What might be wrong with this assumption?
  - Assumes classical bivalence i.e. does not allow for Kleene semantics and # to characterise vagueness
  - Assumes law of the excluded middle and so excludes non-classical logics like Intuitionistic logic

Assumption (ii): properties of ONs

- l denotes an ON while m is a corresponding possibility operator:

$$mp =_{\text{df}} \sim l \sim p$$

- ONs map propositions to propositions
- Williamson assumes 'normality' i.e. each ON commutes with conjunction or $l(\bigwedge_i p_i) = \bigwedge_i lp_i$
- Williamson assumes 4 closure principles:
  1. The truth operator is an ON: $tp =_{\text{df}} p$
  2. composing two ONs yields another ON: $(l_1 \circ l_2)p =_{\text{df}} l_1(l_2 p)$
  3. Any conjunction of ONs is an ON: $(\sqcap_i l_i)p_i =_{\text{df}} \bigwedge_i l_i p_i$
  4. each ON has a 'converse' ON: l' is s.t. p entails $lm'p$ i.e. has always been is converse of will always be
- MN: L is the conjunction of every ON, and we define MN to be L
- Taken together, we reach that the logic of L is at least S5
- What might be wrong with this assumption?
  - Why is the truth operator an ON? In what sense is a proposition true in some world necessary in that world? It seems to cheapen the notion of contingent truths
  - Does composing two ON operators always yields another ON? Take for example 'it is epistemically possible that' and 'it is nomically necessary that' – a conjunction is just 'it is epistemically possible that it is nomically necessary that' which seems quite messy. The point is that Williamson ignores the fact that certain modalities may be in non-overlapping domains such that a combination of the two become meaningless for instance, a modal operator that 'it is nomically possible that it is metaphysically necessary that' is pretty strange
  - Does the conjunction hold? Williamson ignores the facts that combination may result in new things being entailed for instance 'it is necessary by both physics and math (together) that p' may very well not entail 'it is necessary by (a nominalised) physics that p' and 'it is necessary by math that p'
  - Duality assumption is fine

Anti-scepticism about MN:
Williamson focuses on various kinds of scepticism:

- MN is unintelligible (Hume, Quine)
  Hume thinks that the modal operator 'must' expresses a psychological expectation that billard balls bounce off the sides rather than an objective necessity
  Quine thinks that there is only one world
- we have no (non-trivial) knowledge about MN
- MN is non-fundamental/uninteresting (Lewis, Sider)
  Lewis formulates quantified modal logic in terms of concrete worlds and counterpart theory – hence talk about modality *reduces* to talk about maximal connected spatiotemporal systems other than our own.
  Sider thinks that MN is not a unified notion and just a miscvellaneous ragbag of disparate elements from different modalities

Williamson focuses on nomic necessaity and makes the following arguments

- Step 1: if NN is not 'in trouble', neither is MN
- Step 2: NN is in good-standing

Step 1: if NN is not 'in trouble', neither is MN

- Criticism is often focused specifically on metaphysical modality but Williamson thinks most challenges to MN generalise to other ONs

  For instance, the epistemological challenge 'If something is non-actual, how do you know whether it is possible?' arises for any non-trivial objective modality, not just for the metaphysical sort

  Quine's logical qualms about quantifying into the scope of modal operators do not depend on whether those modal operators are interpreted as metaphysical, nomic or practical and Lewis's counterpart theory can be used to interpret other objective modalities too
- So, we need to criticise the whole family of objective modalities
- Williamson thinks that nomic possibility (NN) i.e. consistency with laws of nature, depends on MN and so focuses on it
  - Case i: consistency = logical consistency
    'Hesperus ≠ Phosphorus' is consistent with the laws of nature so it is nomically possible but Williamson thinks this is absurd
  - Case ii: consistency = logical consistency, but conjoin laws of nature with all true claim of identity, kind membership etc
    ...those are just the sorts of move philosophers make in trying to reductively define metaphysical modality itself. It is an illusion one can define a nomic objective modality without running into the issues that beset metaphysical modality
  - Case iii: consistency = metaphysical compossibility
    - NN depends on MN
    - If metaphysical modality is in trouble, so is nomic modality
- In what sense of 'in trouble' does Williamson mean? does it simply mean make sense? Williamson seems to leave this quite vaguely defined or that it means supported by our pre-existing scientific theories

Step 2: NN is in good standing

- If we want to apply a scientific theory freely in the scope about subjunctive reasoning about nomic possibilities, the theory had better be at least nomically necessary. Furthermore, if such modal applications have a track record of success, it provides abductive confirmation for the relevant claims of nomic necessity
- Basically: a scientific thory T supports counterfactuals and consequently, T supports its NN

2 Second-Order Logic

BROAD THEME: Is second-order logic logic?

Response: The following are the differences between FOL and SOL

- SOL includes predicate variables while FOL only includes variables whereby the variables are terms
  - Therefore, for any PC-model $\mathscr{M}$, a variable assignment $g \langle \mathscr{D}, \mathscr{I} \rangle$ assigns a member of the domain to variable terms only in FOL, while in SOL, g also assigns an n-place relation over $\mathscr{D}$ to the n-place predicate variable
- SOL allows for quantification over predicates while FOL only allows for quantification over terms. As a result, SOL has a greater expressive power over FOL and more wffs are valid in SOL than in FOL
- Under standard SOL semantics, compactness fails, the Lowenheim-Skolem theorem fails and Completeness fails

Core Reading
Quine, Philosophy of Logic (2nd ed., Harvard UP, 1986), Ch. 5, 61–70
Quine's argument can be reformulated to be the following:

1. SOL embeds set-theory into its conception of 'logic'
2. Set theory is not logic
3. Therefore, SOL is not logic

On what grounds does Quine justify (i)?

- Quine draws an analogy between ordinary quantification in FOL and name positions – when we say $\forall x$(x walks), we treat x as a name position such as 'Matthew'.

  Likewise, when we quantify over predicates, we treat the predicate as name positions as well for some sort of entity.
- Logicians then treat predicate letters as sets because they want to find a way to establish the sameness of predicates, which follows from the axiom of extensionality which holds that two sets are equivalent iff their extensions are the same; attributes do not have this property – animals which have hearts might have the same extension as the attribute of animals having kidneys, but they nonetheless aren't the same attribute.

  And if we treat predicate variables as name positions for sets, then we smuggle in truths of set theory such as $\exists G \forall x(Gx \leftrightarrow Fx)$ as logic validities since this follows from the SOL tautology $\forall x(Fx \leftrightarrow Fx)$.

Just because SOL includes more logical truths than FOL is not ipso facto a reason to deny SOL the status of logic. Quine therefore needs to prove that (ii) is true to reach the conclusion expressed in (iii).

On what grounds does Quine justify (ii)?

- I think his argument here is less clear.

  He argues that "We can in this way enjoy the convenience of an ontology of sets...without footing the ontological bill" but that there will be an "ontological reckoning". It is not clear what this ontological reckoning is.

- However, I take Quine's argument against Set Theory being logic as a pragmatic argument. Logic is supposed to be neutral of theory – even set theory. Therefore, it must have the lowest ontological commitment possible. Quine therefore argues that we need not use predicate variables to admit sets as "values of quantified variables", and can instead use the extralogical vocabulary of "$x \in y$" to resolve this need. We can define the membership criteria of a set in FOL without appealing to SOL.

- Crucially, I think that we can add the fact that to include set theoretical truths as logical validities is to presuppose a commitment to a certain theory (ZFC in this case) and their accompanying axioms which might not be logical tautologies.

- In short, Set Theory is not logic because logic is supposed to be theory-neutral, free of ontological and theoretical commitments that Set Theory makes.

- Logic is a theory-neutral language to express our arguments to ensure truth-preservation in virtue of the argument's form and structure. In this sense, because SOL is motivated to help reduce mathematics into logic, it invariably adds extra-logical notions and commitments to FOL, rendering it no longer logic as we conceive of it.

Boolos, On Second-Order Logic, Journal of Philosophy 72 (1975), 509–527. Reprinted in his Logic, Logic and Logic.
Boolos's argument mainly seeks to refute Quine's own argument against SOL being considered logic.
Basic position: Quine is wrong to think that SOL implies that predicates take name positions

- Boolos argues that, contra Quine, quantifying over predicates merely means that a predicate F has a range, rather than being a name position.

- Furthermore, he argues that SOL does not in fact have many ontological commitments – the formula $\exists X \exists x \exists y (Xx \wedge Xy \wedge \sim x = y)$ is not valid in SOL and so SOL is not committed to the existence of a two-membered set.

- I think his criticisms of Quine's argument goes wrong here.

- First, it is difficult to see why just because a predicate F has a range means that it is not a name position. Consider the analogy of $\exists x$(x walks). Here, we see that the variable term x also has a range (perhaps of people), but this nonetheless does not defeat the fact that it is a name position because we substitute x for names of various people.

- Second, I think Boolos misattributes the force of Quine's objection to SOL's ontological commitments. It is not the mere fact that SOL has vast ontological commitments to the existences of sets, but rather the mere fact that it is committed to certain types of entities, when logic is supposed to be theory-neutral with as little commitments as possible.

Boolos thinks that set theory is distinctively logical

- Boolos's reply to the second objection I made is as such: he thinks that notions of set, class, property, concept, and relation have been considered to be distinctively logical notions.

- This is because one can say that a set itself is topic-neutral – anything can belong to a set, have a property or be related to something. So, if one's necessary criterion for something to be logic is that it is theory neutral, and if Boolos is right, then he seems to have made some progress favouring SOL as logic.

- But I do not think that this is an effective reply. While anything might be included in a set, such as Socrates or numbers, the problem is that what a set is entails certain kinds of properties i.e. the axiom of choice or the axiom of extensionality, which may not hold for other kinds of entities such as attributes.

  For instance, the sets of animals who have hearts and animals who have kidneys are the same set by the axiom of extensionality, but are clearly not the same attribute. SOL therefore commits itself to a certain kind of (non-logical) structure and assumptions working in the background. This is why Boolos' reply fails, and why his general argument fails.

Further Reading
Shapiro, Foundations without Foundationalism, chs. 4, 5 and 7
Chapter 4: Metatheory

- Basically, we need to use Henkin semantics to restore the properties of FOL in SOL (which fail under standard semantics)

  Standard semantics: Quantifiers over predicates range over all possible subsets (or relations/functions) over the domain. So if your domain is D, then a second-order quantifier over unary predicates ranges over all subsets of D

  Henkin Semantics: Second-order quantifiers range over a specified collection of relations/functions — not necessarily all possible ones.

- However, Henkin Semantics loses the advantage of Standard Semantics in being unable to express categoricity

  A theory is categorical if it uniquely determines the structure it is meant to describe — up to isomorphism.

  For SOL with standard semantics, That is, there is only one model of the axioms — the standard natural numbers $\mathbb{N}$ — up to isomorphism.

- Henkin semantics restores the Downward Lowenheim-Skolem Theorem, Compactness and Completeness.

  Peano axioms become non-categorical: Due to the Löwenheim–Skolem Theorem and Compactness, Peano Arithmetic has non-standard models (with "infinite integers" or weird elements)

Chapter 5: SOL and Math

- Shapiro says that Quine is right in that SOL can express a great deal of mathematics and they have substantial mathematical and ontological presuppositions and that SOL is thoroughly intertwined with set theory

- But he denies that these presuppositions are too great for SOL to bear the label of 'logic'

  If there were a sharp border between logic and mathematics, SOL would go on the mathematics side; but Shapiro thinks that there is no natural boundary to be dawn

  From the Quinean 'web of belief' perspective, Quine himself has argued that mathematics is necessary for understanding just about anything – why would logic, especially the logic of mathematics, be different? And if the logic of mathematics is required to understand just about anything, then this 'topic-neutrality' that Quine alludes to is achieved (this is Boolos' argument)

Chapter 7: Against the monopoly of FOL over the label of logic

- Different schools of thought: The logicist camp, which included Russell and Frege, wanted to capture mathematics in terms of logic and hence advocated for HOLs; the axiomatic camp, which included Dedekin, Peano and Hilbert

- Important consequence of the Lowenheim-Skolen Theorem is that no first order theory with an infinite model is categorical (hence no unique candidate can be singled out by the axioms of set theory) – but this presupposes FOL since LS is a property of only FOL (and SOL with Henkin Semantics)

- Zermelo vs Godel debate: Godel's insistence on first order finitary languages is because Godel thought that logic should focus solely on the formal. That is, derivations within deductive systems should be checkable and hence what must emerge is *compactness* and *completeness*.

  Zermelo, after hearing about the incompleteness of any recursive axiomatisation of arithmetic, thought that compactness was an unacceptable property of logic because we should expand our notion of proof. He thought that Godel held an overly restrictive notion of proof.

- The standoff currently mirrors the historical standoff between FOL and HOLs: whether the 'logical notion of the set' is sufficiently clear/topic-neutral for us to use in logic and whether the use of such terminology is essential to at least part of the foundational enterprise.

  Alternative is to use the non-logical operator of set membership

- Quine holds that there is no sharp border between logic/mathematics and natural science.

  His Quine-Putnam indispensability argument shows how mathematical concepts, involving a commitment to its otnology, are essential in virtually all sciences – anti-nominalism

  Yet, he inconsistently believe that logic – the logic of mathematics – is different

3 Existence and Necessity

BROAD THEME: Is everything necessarily something?

Core Reading
Williamson, Bare Possibilia, Erkenntnis 48 (1998), 257–273.
Basic position: Williamson validates the Barcan formula and thus SQML via committing to the existence of *bare possibilia* i.e. things that exist as mere possibilities and not concretely.
The BF and CBF and its counterexamples:

- The Barcan Formula is provable in SQML

  $$\vdash_{\text{SQML}} \forall x \Box \phi \rightarrow \Box \forall x \phi \tag{BF}$$

  By the soundness of SQML, we have that BF is valid in SQML,

  $$\models_{\text{SQML}} \forall x \Box \phi \rightarrow \Box \forall x \phi$$

  Similarly with the Converse Barcan formula, we have

  $$\vdash_{\text{SQML}} \Box \forall x \phi \rightarrow \forall x \Box \phi \tag{BCF}$$

  and by soundness,

  $$\models_{\text{SQML}} \Box \forall x \phi \rightarrow \forall x \Box \phi$$

- Given that BFC and BF are theorems of SQML, they face no possible counterexamples i.e. there is no model $\mathscr{M} = \langle \mathscr{D}, \mathscr{I}, \mathscr{W} \rangle$ in which BF or BCF are false.

- But represented in English, they seem to be vulnerable to *prima facie* counterexamples. The BF essentially states (in its logically equivalent existential form) that if there could have been an x that is a ϕ, then there is some x that could have been a ϕ. But this seems to be intuitively false.

Suppose we take ϕ as 'Wittgenstein fathered x' and although Wittgenstein did not have children, it is plausible that he possibly could have had a child.
But the consequent of BF commits us to picking out an object that could have been fathered by Wittgenstein. And since it seems that your parentage is essential to you as a person, no actual object (person or non-person) that exists in our world could have been fathered by Wittgenstein.

- The BCF also succumbs to similar counterexamples, though for our purposes I will just focus on BF since BFC can be derived from BF.

Williamson's solution

- Williamson's solution to disarming these purported counterexamples is to commit to the existence of *bare possibilia* or objects that merely exist as possibilities and not as concrete objects.
- In short, Williamson is committed to the necessity of existence: $\Box\forall x\Box\exists y(x = y)$, and says that in every world w, everything exists.
  hat matters is in what sense they exist – either concretely or as simply *bare possibilia* with no spatiotemporal location.
- Therefore, the candidate counterexamples to BF and BCF do not hold (Wittgenstein's child exists with no spatiotemporal location), and are instead counterexample to a different claim that if there is something in space and time that could have been ϕ, then there could have been something in space and time that was ϕ.
- Problem? Such an ontological commitment to the necessity of everything existing, but in different modes seems costly.

  Furthermore, Williamson must justify this commitment over the alternative to relativise domains to worlds i.e. that $\mathscr{D}(w_1) \neq \mathscr{D}(w_2)$.
- Williamson's argument falters here: he argues that the relativisation of domains to worlds creates great complications for formal semantics and that completeness proofs are "more convoluted". But this is not ipso facto a reason to choose SQML over VDQML, and sacrifice our commonsense intuitions about what exists.
- Williamson also smuggles in a controversial notion of existing – clearly, what is up for debate is whether bare possibilia do exist in the way that we think $\exists x$ captures.

  Of course, we do not think that having a spatio-temporal location is necessary to exist (intuitively, numbers exist for instance), but to think that even possibilia without names to refer to them exist seems to do an injustice towards the notion of existence.

  When I say 'Matthew exists', I would not be happy with Matthew existing as possibilia, but an actual Matthew existing in my world. Williamson's solution has not actually resolved the problem, merely delayed it by making the problem about what counts as existence.

Hayaki, Contingent Objects and the Barcan Formula, Erkenntnis 64 (2006), 75–83
Basic Thesis: The cost of adopting the ontology of bare possibilia is higher than it first appears, and higher than the benefit of a simpler logic.

Williamson/Linsky/Zalta's argument:

- The argument against BF is: although Wittgenstein died childless, there could have existed an object fathered by him. By BF, it would follow that there actually exists an object that could have been fathered by Wittgenstein. But no actual object was fathered by Wittgenstein, so if we accept the doctrine of the necessity of origin, we must reject BF

- Response: While there is no actual person that could have been fathered by Wittgenstein, there is nevertheless an actual object that could've been fathered by Wittgenstein – does not occupy space-time in this world and is 'contingently non-concrete'.

  In another world, this object is concrete and is a person fathered by Wittgenstein.

Hayaki's Response: highly counter-intuitive response

- What we usually think of as non-existence (non-actuality) has been relabelled as non-concreteness – we push the boundary or threshold for existence back.

  Distinction between existence and non-existence has been rendered worthless.

- Instead of 'George W. Bush exists in some worlds, but not in others', we instead mean to say 'George W. Bush exists in all worlds, but is concrete in some and not others'
- A deeper problem is now, we cannot say that George W. Bush is essentially concrete since it would be (since we have to replace 'exists' with 'is concrete')

  $$\Box(\text{George W. Bush is concrete} \rightarrow \text{George W. Bush is concrete})$$

  But this is vacuously satisfied by any object, not just George W. Bush, and not all objects are essentially concrete

- Potential Response: Linsky and Zalta have to deny that George W. Bush is essentially concrete and they would also have to deny that Bush is essentially human, on the grounds that he fails to be human (as he fails to be concrete) in some world.

  This commitment to anti-essentialism bankrupts any meaningful talk of essences as they deny a paradigmatic case of essentiality.

- Potential Response: they can concede that George W. Bush's essence does not consist in being concrete, or human, or (minimally) rational, but in being *possibly* concrete, *possibly* human etc.

  But then, we cannot express the distinction between objects that have F essentally, and those that have F contingently; both come out as essentially possibly F

- Potential Response: Make the metaphysical assumption that no object can straddle the concrete/abstract division: no object can be concrete in one world and abstract in another, although it can be concrete in some worlds and non-concrete in others. The essential concreteness of Bush could be preserved by redefining 'essentially concrete' to mean 'possibly concrete' (and necessarily not abstract).
- Further problem with fictional characters: We usually regard fictional characters as being created (say Sherlock Homes by Arthur Conan Dolye). But if Sherlock Holmes were a necessary existent, then Conan Dolye could not have created it, merely discovered Sherlock Holmes.

- Another problem is how we might deal with the existence of abstract objects: there seems to be more abstract objects than there actually are – Consider a possible world in which Harry Potter does not exist. On Linsky and Zalta's view, these abstract entities do exist at that possible world (since they exist everywhere) but cannot be described as non-concrete, since that implies that they are concrete elsewhere. Can they be said to be 'contingently non-abstract', on par with non-concrete objects which are concrete at other worlds? But they are surely abstract if they exist; they certain are neither concrete nor non-concrete.

  Have to invent new terminology like 'contingently unrealised'

- Formalisation issues: translations for 'x is essentially F' will double when we take abstracta into account:

  1. 'x is essentially human' will be '$\Box x$(is concrete $\rightarrow$ x is human)' but this formula must be restricted for use only with possibly concrete x. Since the number 17 is concrete with no worlds, it would come out as being essentially human. Thus a more accurate translation would be '$\Diamond$(x is concrete $\wedge$ $\Box$(x is concrete $\rightarrow$ x is human)'. This assumes that no object can straddle the concrete/abstract divide
  2. 'x is essentially concrete' will either be $\Box$(x is concrete)' and false for all x or $\Diamond$(x is concrete)' and is true for all x that are concrete in at least one world
  3. 'x is essentially fictional' will be '$\Box$(x is realised (?) $\rightarrow$ x is fictional)' but again this formula must be restricted for use only with possibly abstract x

  This replaces the single format required for more standard ontology: 'x is essentially F' is usually translated as '$\Box$(x exists $\rightarrow$ x is F)'

  Parsimony dictates we reject fixed domains across worlds.

- Williamson's additional argument: Suppose we have two knife blades and two handles, B_1, B_2, H_1, H_2 and we ask 'How many knifes could have been made from them' – in one answer, 2 and another sense 4. Williamson takes this to mean that in this world, there are 4 possible knifes but no more than 2 actual knifes

  Remark: But isn't possible meaning there in some other world, there could have been some other combination of the blades and the slotted handle? Are we not counting across worlds?

- Hayaki's response: are we really counting possible objects and whether we count actual objects

  Possible objects nor ways of fitting blades are as ontologically respectable as one might wish – but we nevertheless routinely quantify over ways (more than one way to skin your cat), sometimes more naturally than we quantify over possible objects.

  But this does not settle the question of their actuality – we can quantify over possible objects, but on the counterpart-theoretic semantics of Lewis, we can argue that no object exists in more than one possible world. Possible knifes are thus not actual objects

Further Reading
Tim Williamson, Modal Logic as Metaphysics (OUP, 2013), chs 1, 2.1-2, 3, 4.1
Chapter 1: Overview of Necessitism vs Contingentism

- Necessitism is the claim that necessarily everything is necessarily something

  Absolutely universal – about objects, not about kinds. It is contingent that *animals as a kind* exist, but an *animal* would have been something else otherwise. Or, the Sun would necessarily be something even if there were no stars.

- Necessitism does not imply 'The F is an F' i.e. necessitism does not imply that there is a golden mountain and thus that the golden mountain is a golden mountain. It merely implies that there is a merely possible golden mountain, but only on the attributive reading: something could have been a golden mountain, but is not a golden mountain.

- Dissatisfaction with actualism and possibilism: what is it for something to be actual or to be possible? "To be actual is to be in the actual world" does not help

Chapter 3: Possible Worlds Model Theory

- Williamson aims to show how, in particular, the model theory of QML bears on a substantive, if highly abstract, dispute in modal metaphysics and that QML is not *neutral*

- Williamson's strategy: Standardly, it is thought that we can bring model theory to bear on metaphysics by isolating an intended class of models. Models in the intended class interpret the expressions of the language of QML in the way we intend when we are asking after the features of modal reality. So, the sentences verified by all models in such a class may reasonably be taken accurately to characterize features of modal reality. But which classes of models are intended?

  Williamson's strategy for answering this question proceeds in two steps

  1. First, isolate what one might plausibly take to be the logical truths concerning metaphysical necessity: a set of formulae in QML that $\Box$ is interpreted as metaphysical necessity
  2. Second, specify a class of intended model structures by appeal to the set of logical truths: a structure is intended iff its logic – the formulae verified by all models with that structure – coincides exactly with the set of logical truths concerning metaphysical necessity. Then a class of models is intended if it is the class of models of an intended structure

Chapter 4: Argument from the Being Constraint

- The Being constraint: having properties requires existence

$$\Box\forall x\Box(Fx \to \exists z\ x = z)$$

  How could a thing be propertied were there no such thing to be propertied? How could one thing be related to another were there no such things to be related? (Williamson, 2013, p. 148).

  Model-theoretically, the being constraint corresponds to the domain constraint: predicate extensions at a world must be drawn from that world's domain.

- The contingentist must reject the Being constraint since

  1. Logic ensures the necessity of self-identity: $\Box\forall x\Box x = x$
  2. For reductio, assume $\Box\forall x\Box(Fx \to \exists z\ x = z)$
  3. (i) and (ii) imply $\Box\forall x\Box\exists z\ x = z$

  So, the contingentist needs to reject (ii), but (ii) is an instance of the Being constraint.

- Williamson: Contingentists who accept (ii) should, TW says, accept the being constraint in general. Self-identity is the easiest property to have; so if you can't even have that without existing, you surely can't have any properties at all without existing.

  So, the contingentist cannot say they accept (ii) without accepting the Being Constraint

- Contingentists are in a tricky position. If they insist that it is possible to fall under a predicate and yet be nothing, they face the charge that they are unserious about their own contingentism, because they are tacitly restricting the quantifier 'nothing' (section 4.1). If they agree that falling under a predicate entails being something, they slide into necessitism unless they distinguish not falling under a predicate from falling under a negative predicate, which is best done by means of something like the λ operator. If they introduce the λ operator, they still slide into necessitism unless they complicate its logic in awkward ways. Although none of this amounts to a refutation of contingentism, it is evidence that the view goes against the logical grain. (Williamson, 2013, p. 188)

Sider, Williamson's Many Necessary Existents, Analysis 69 (2009), 250–258.
Basic position: Sider's argument runs as an argument by reductio.

1. If we accept an iterative conception of a set, there is a set of all nonsets

2. Lewis' modal realism licences a principle of recombination
3. Modal Realism applies to Williamson's view about many necessary existents
4. The recombination principle, together with BF and BCF (in its infinitary form), entail that if we attempt to construct a set of all nonsets with cardinality v, we can construct a set with cardinality greater than v that contains only nonsets, and thus contradicting our assumption that there is a set of all nonsets
5. Hence, we should not accept Williamson's argument for accepting BF and BCF, and should reject them

My basic objection: Disagree with Sider. I am going to challenge premise (3) and the assumptions that allow for premise (4) to hold true. In particular, the tenor of my argument is just because Lewis's modal realism and the auxillary assumptions Sider makes are consistent with Williamson's argument, does not mean that Williamson's argument entails them or requires them.
Sider takes Williamson to assume the following principles to prove his argument

- Since by Sider's admission that these assumptions are not strictly required by Williamson's argument, we can freely assess whether to reject them
- Specifically, I suggest that the principle of the necessity of identity is not at all obvious. An intuitive counterexample would be more easily generated once we look at its consequence of the necessity of distinctness
$$\Box\forall x\forall y(x \neq y \rightarrow \Box x \neq y) \tag{ND}$$
Consider: Some table, call it Woody, made out of material w is obviously distinct from some other table made out of material m. Yet, it is by no means necessary that Woody has to be made out of material w, and thus can be made out of material m (say m is some material similar to w). So, there is some possible world in which $x = y$, meaning $\Box x \neq y$ is false, and the principle of ND to be false. More easily, it seems plausible that there is some possible world in which water and H2O are distinct, rendering NI false. So, if Sider's premise 4 relies on NI and ND, then his argument runs into problems, for Williamson has no reason to accept NI.
- I also suggest that while modal realism might apply to Williamson's argument for accepting the Barcan Formula, it is not true that Williamson's argument requires modal realism.
The intuition runs like this: Lewis's modal realism (about the way possible worlds are) is quite independent from the kinds of logic (and accessibility relations between worlds) we endorse.
- Lewis's modal realism requires the principle of recombination because he intends to define possible worlds without the use of primitive modality and instead as a maximal spatiotemporally interrelated whole.
But should Williamson instead endorse Stalnaker's moderate realism or some sort of ersatz realism, he does not require the principle of recombination that is the key piece of Sider's argument.
- It might very well be the case that Williamson is a modal realist, but it seems clear to me that talk about the systems of modal logical we should accept to describe modality runs quite independently from the ways in which possible worlds are. So, modal realism has little bearing on Williamson's many necessary existents, and Williamson can reject the principle of recombination in favour of a theory of possible worlds that accepts primitive modality (perhaps Rosen's fictionalism). In short, we can deny premise 3.

4 The contingent a priori

BROAD THEME: What is the nature of the contingent a priori?

Core Reading
Ted Sider, Logic for Philosophy, ch 10.4

Evans, Reference and Contingency, The Monist 62 (1979), 161–89

Martin Davies, Reference, Contingency, and the Two-Dimensional Framework Philo- sophical Studies 118 (2004), 83–131. Sections 1, 3 and 4.

5 Conditionals

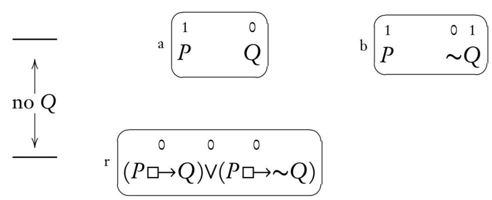
BROAD THEME: What is the correct formalisation of conditionals?

Core Reading
Ted Sider, Logic for Philosophy, ch 8.7

- Stalnaker's System and what he licenses:

$$\vDash_{SC} (\phi \Box\!\!\rightarrow \psi) \vee (\phi \Box\!\!\rightarrow \sim \psi) \qquad \text{("Conditional excluded middle")}$$
$$\vDash_{SC} [\phi \Box\!\!\rightarrow (\psi \vee \chi)] \rightarrow [(\phi \Box\!\!\rightarrow \psi) \vee (\phi \Box\!\!\rightarrow \chi)] \qquad \text{("distribution")}$$

- Against Conditional Excluded Middle: Consider



Here, worlds a, b are tied for similiarty to r.

An equivalent formulation of the conditional excluded middle is

$$\sim (\phi \Box\!\!\rightarrow \psi) \rightarrow (\phi \Box\!\!\rightarrow \sim \psi)$$

which is controversial; the converse is agreed by everyone to be true. So this results in us thinking that $\sim (\phi \Box\!\!\rightarrow \psi)$ and $(\phi \Box\!\!\rightarrow \sim \psi)$ are equivalent. But we don't often do this. Consider

  - 'it is not true that if she played, she would have won'
  - 'if she had played, she would have lost'

Clearly they are not equivalent formulations because there might be equally close worlds in which she played but she won in one but lost in another.

- Against distribution: If someone says 'If I had been a baseball player, I would have been either a third-baseman or a shortstop', it might seem natural to reply with a question 'well, which would you have been?' which presupposes either 'if you had been a baseball player, you would have been a third-baseman' or 'if you had been a baseball player, you would have been a shortstop' must be true.

- Two types of denial:

  Metaphysical: If we accept the notion of comparative similarity, which allows for ties, we must give up antisymmetry.

Semantic: The intuitions for each aren't completely compelling.
For the baseball case, consider a parallel with coin flipping – it is definitely true that if the coin had been flipped, it would either be a heads or tails. But this only implies semantically that it might be tails or it might be heads, not that either, it would be heads or it would be tails.

- Problem of limit assumption: Lewis asks us to consider the example "If the line (that is currently less than one inch long) had been longer than one inch, it would have been one hundred miles long".

  If we use Stalnaker's truth conditions for natural language counterfactuals, and take our intuitive judgements of similarity at face value, the sentence comes out as true, for there does not seem to be a closest world in which the line is more than one inch long – for every world which is $1 + k$ inches long, there is a closer (more similar) world where the line is $1 + \frac{k}{2}$ i.e. a sequence of ever-closer worlds would converge back to the real world

Lewis, Counterfactuals (Blackwell, 1973), 3.4, 'Stalnaker's theory'.

- Stalnaker's theory: $\phi \mathrel{\Box\!\!\rightarrow} \psi$ is true at a world i iff either (1) ϕ is true at no world accessible from i (vacuous case) or (2) ψ is true at the ϕ-world closest to i
- Stalnaker's assumptions
  1. The Limit Assumption: that there never are closer and closer ϕ-worlds to i without end (so there is some closest ϕ-world)
  2. The antisymmetry assumption: if there is a ϕ-world, then there is exactly one closest ϕ-world

     Remark: Lewis thinks that this assumption is stronger than the limit assumption

  We can think of Stalnaker's theory in terms of a selection function f.

  Definition 5.1 (Stalnaker's selection function). Given any antecedent ϕ and world i such that some ϕ-world is accessible from i, f picks out a single world $f(\phi, i)$: one of the ϕ-worlds accessible from i regarded as the one closest to i

  Remark: we have two implicit constraints on f:

  1. Whether i itself is a ϕ-world, $f(\phi, i) = i$
  2. Whether ψ holds at $f(\phi, i)$ and ϕ holds at $f(\psi, i)$, $f(\phi, i)$ and $f(\psi, i)$ are the same world

- Lewis (rightly) argues that Stalnaker's theory is a special case of his where his selection function f_L is a set-selection function (since it gives up anti-symmetry)
- Problems with Stalnaker's theory:
  - Fundamentally, Lewis thinks that Stalnaker's theory licensing the Law of the Conditional Excluded Middle:
  $$\models_{SC} (\phi \mathrel{\Box\!\!\rightarrow} \psi) \vee (\phi \mathrel{\Box\!\!\rightarrow} \sim \psi)$$
  which does not hold in Lewis's system as a bad thing, despite its initial plausibility
  - Selection forced to choose
  The plausibility of the Law of Conditional Excluded Middle rests on the fact that we do not distinguish, in ordinary language, between $\sim (\phi \mathrel{\Box\!\!\rightarrow} \psi)$ and $\phi \mathrel{\Box\!\!\rightarrow} \sim \psi$.
  Except in the vacuous case, $\phi \mathrel{\Box\!\!\rightarrow} \sim \psi$ implies $\sim (\phi \mathrel{\Box\!\!\rightarrow} \psi)$ on both Lewis's and Stalnaker's theory; given the Law of the Excluded Middle, the former, $\sim (\phi \mathrel{\Box\!\!\rightarrow} \psi)$, also implies that latter, $\phi \mathrel{\Box\!\!\rightarrow} \sim \psi$.

    Lewis says the Conditional Excluded Middle prohibits us from truly saying that:

It is not the case that if Bizet and Verdi were compatriots, Bizet would be Italian; and it is not the case that if Bizet and Verdi were compatriots, Bizel would not be Italian; nevertheless, if Bizel and Verdi were compatriots, Bizel either would or would not be Italian

i.e. $\sim(\phi \mathrel{\Box\!\!\rightarrow} \psi) \wedge \sim(\phi \mathrel{\Box\!\!\rightarrow} \sim\psi) \wedge (\phi \mathrel{\Box\!\!\rightarrow} \psi \vee \sim\psi)$.

This doesn't work in Stalnaker's system because the first two sentences requires it to be the case that in the closest ϕ world, Bizet is not Italian, and that in the closest ϕ world, Bizet is Italian, and that there can only be one closest ϕ world bearing these incompatible properties

The problem, according to Lewis, is that however little there is to choose for closeness between worlds where Bizet and Verdi are compatriots for both being Italian and worlds where they are compatriots by both being French, the selection must still choose under Stalnaker's system. But Lewis denies that the selection function can choose, or at least not based off comparative similarity because comparative similarity allows for ties (Berdi is actually French while Verdi is actually Italian). The point is that you can't really have a linear order on worlds.

– Loss of distinction between 'would' and 'might' due to Law of Conditional Excluded Middle.

Definition 5.2 (Might). Lewis defines the might operator as

$$\phi \mathrel{\Diamond\!\!\rightarrow} \psi =_{\text{df}} \sim(\phi \mathrel{\Box\!\!\rightarrow} \sim\psi)$$

As a consequence of the law of the excluded middle, $\phi \mathrel{\Diamond\!\!\rightarrow} \psi$ implies $\phi \mathrel{\Box\!\!\rightarrow} \psi$; and $\phi \mathrel{\Box\!\!\rightarrow} \psi$ implies $\phi \mathrel{\Diamond\!\!\rightarrow} \psi$ except in the vacuous case (the second clause holds for both LC and SC).

But how can $\phi \mathrel{\Box\!\!\rightarrow} \psi$ and $\phi \mathrel{\Diamond\!\!\rightarrow} \psi$ not differ in truth value except in the vacuous case?
Stalnaker could try to define 'might' in some other way but Lewis denies that any works. Consider

1. $\Diamond(\phi \wedge \psi)$
2. $\Diamond(\phi \mathrel{\Box\!\!\rightarrow} \psi)$
3. $\phi \mathrel{\Box\!\!\rightarrow} \Diamond\psi$
4. $\phi \mathrel{\Box\!\!\rightarrow} \Diamond(\phi \wedge \psi)$

But none work because consider ϕ is 'I looked in my pocket' and ψ is 'I found a penny': suppose I did not look and suppose there was no penny to be found, and make commonplace assumptions about relevant matters of fact. Then 'If I had looked, I might have found a penny' is plainly false, but all four candidate symbolisations are true

- Lewis suggests that the obvious revision of Stalnaker's theory is that there might be more than one equally closest antecedent world – via set selection functions f_L.

  This only removes antisymmetry and retains the limit assumption, but Lewis thinks that the limit assumption is safer than antisymmetry – I agree.

  Remark: the limit assumption merely imposes the restriction that there are no uncountably infinite worlds

- Another way is to simply pick an arbitrary ϕ-world from the set of closest worlds – thereby denying that we are determining the choice of worlds by comparative similarity but simply picking some admissible world.

  Obviously this makes it arbitrary

Stalnaker, Inquiry (Cambridge University Press, 1984), ch. 7 , 'Conditional propositions'.

- Standard explication of selection function and motivation for rethinking 'if A had happened, then B would occur' as $\mathrel{\Box\!\!\rightarrow}$ instead of $\rightarrow$. Consider:

1. Against transitivity:
   "If the Democrats has lost control of the House, the Republicans would be in control;
   if the Socialist Labour Party had gained a majority in the House, the Democrats would have lost control;
   therefore, if the Socialist Labour Party had gained a majority in the House, the Republicans would be in control"
2. Against contraposition:
   "My dog is a mutt. His paternity is in some doubt, but even if his father were a purebred dog, my dog would still be a mutt since his mother was one. If my dog were purebred, his father would be a mutt."

- Criticisms of Stalnaker's use of Comparative Similarity:
  - Too vague and empty to determine truth-conditions for conditions: What does it mean for Seattle to be more similar to San Fransico than it is like Los Angeles?

    Lewis thinks this is actually a strength because we can make sense of comparative similarity judgments about large and complex entities
  - Counterexamples to the hypothesis that a counterfactual is true iff its consequent is true in the most similar world in which the antecedent is true.

    Counterexamples try to show: counterfactuals which intuitively seem to be obviously true even through the consequent was false in all the possible worlds that seemed, intuitively, to be overall the most similar worlds in which the antecedent was true.

    Fine: "If Nixon pushes the button, there would have been a nuclear war" This seems true, given the right context, even though some possible worlds in which Nixon pushes the button and there is no war seem much more similar to the actual world.
    Consider a world in which after Nixon pushes the button something intervenes – perhaps in this world, a switch is defective and the message fails to get through, and then Nixon changes his mind and things go on pretty much as they actually did.

    It does not seem that the fact that such a world is more similar to the actual world, overall, than one in which there is a nuclear war is a good reason for saying that these things would have, or even might have, happened.

    Stalnaker: this example shows decisively that the intuitive notion of overall similarity between possible worlds does have some content and that this notion is not the one that is relevant to the interpretation of counterfactual conditionals.

    Not overall similarity but some other kind of similarity.

    Tichy: No kind of similarity plays a part in closest-world selection

    Suppose there is a man who always wears his hat when it rains, but when it does not rain, it is a matter of chance whether he wears his hat or not. Today, it rained and he wore his hat. What in this situation, should we say about the counterfactual, "if it hadn't rained, the man would have worn his hat"? According to the similarity analysis, no matter how the different respects of similarity are weighted, the counterfactual comes out true. In choosing between possible worlds in which it rains and by chance the man does not, there is no question of trade-offs between different respects of similarity. One can choose a world in which the man wears his hat, as he does in the actual world, without giving up any respects of similarity at all

But this obviously wrong: if it is really a matter of chance whether the man wears his hat when it doesn't rain, then we cannot say whether he would have or not this time if it hadn't rained.

Stalnaker: Goodman taught us that counterexamples do not threaten the abstract. The motivation for the doctrine that the selected world is in some sense minimally different from the actual world is that it seems necessary to explain how evidence from the actual world is relevant in the way it is to the truth of the counterfactual.

– Assumptions that Stalnaker imposes on the selection function – the limit assumption and the anti-symmetry. Stalnaker argues that one is reasonable to be made (limit) and the other need not be made in application (uniqueness)

  * Limit assumption entails the following consequence condition for conditionals: for any set of of propositions Γ and propositions A and C, if Γ semantically entails C, then $\{A \mathrel{\Box\!\!\rightarrow} B : B \in \Gamma\}$ semantically entails $A \mathrel{\Box\!\!\rightarrow} C$
  * To accept in addition the uniqueness assumption is to accept the validity of the principle of the conditional excluded middle.

  Stalnaker's solution: by using supervaluation, we can reconcile the uniqueness assumption, as an assumption of the abstract semantics for conditionals, with the fact that it is unrealistic to assume that our conceptual resources are capable of well ordering the possible worlds.

  Supervaluationism basically says that any partially defined semantic interpretation will correspond to a set of completely defined interpretation – the set of all ways of arbitrarily completing it. A sentence is true in a supervaluation iff it is true in all corresponding classical valuations; a sentence is false in a supervaluation iff it is false in all corresponding classical valuations; a sentence is neither true nor false if it is true on some of the classical valuations and false on others.

  Here, the selection function that is actually used in making and interpretating counterfactuals correspond to orderings of possible worlds that admit ties and incompatibilities.

  Remark: But Lewis isn't saying that we cannot admit ties just because of vagueness – we can also be equally different in reality. Consider 3 is as far away from 4 as 5 is from 4 – which are we to choose?

  Stalnaker says that supervationalism has very little effect on the logic of conditionals – classical logical truths are true in all classical valuations and hence will be true in all classical valuations determined by any partial interpretation and thus be super-true. Also, the law of the conditional excluded middle remains valid when truth and validity are defined in terms of supervaluations even when there may be cases where neither $\phi \mathrel{\Box\!\!\rightarrow} \psi$ nor $\phi \mathrel{\Box\!\!\rightarrow}\sim \psi$ are true.
  Remark: But clearly, most of the cases we consider are those that are vague.
  Stalnaker's Response to the Bizet and Verdi example by Lewis: Quine takes for granted, by using the distribution principle of inference in the example, that a counterfactual antecedent purports to represent a unique, determinate counterfactual situation. It is because counterfactual antecedents purport to represent unique possible situations that examples which show that they may fail to do so are a problem. But one should respond to the problem not by changing the truth-conditions for conditions so that it does not arise, but by recognising that in application, there is great potential for indeterminacy in the truth-conditions for counterfactuals.

  The reason why the distribution principle seems to fail on Lewis's account is because he wrongly thinks that the antecedents of the conditions act like necessity operators on their consequents To assert thar 'If A had happened, then B' is to assert B is true in every one of a set of possible worlds defined relative to A.

Stalnaker thinks Lewis's analysis is contingent on a scope ambiguity in the following case:
X: President Carter would have appoointed a woman to the Supreme Court last year if there had been a vacancy.
Y: Who do you think he would have appointed?
X: He wouldn't have appointed any particular woman; he just would have appointed some woman or other OR I don't know, I just know it's a woman that he would have appointed

If Lewis is correct, then we would think that Y possibly misreads X's statement; but Stalnaker thinks that there is no possible reading on which Y misreads the statement i.e. Y interprets "a woman" to have a wide scope while X might intend for the quantified expression "a woman" to have a narrow scope

Remark: Really? I can see that the alternative response would make it much better

- Two further criticisms by Lewis
  - Supervaluation's success depends on Limit assumption

    Stalnaker think we should just accept the Limit assumption – there is in fact a nonempty set of closest worlds in which A is true.
    The selection function may ignore respects of similarity which are not relevant to the context in which the conditional statement is made.

    Stalnaker says, against Lewis's one inch example, that the selection function will be undefined for such antecedents in such contexts since we cannot use the definite description 'the shortest line longer than one inch' (doesn't exist given the density of $\mathbb{Q}$ in $\mathbb{R}$

    On Lewis's analysis, applied to his example of the line, every conditional of the following form is true: if the line had been more than one inch long, it would not have been x inches long, where x is any real number. This implies, given Lewis's might counterfactual, that there is no length such that the line might have had that length if it had been more than one inch long. But the line, if longer than one inch, must have been some length or other.
  - Still no Might conditional: Stalnaker admits that if what Lewis says is true, that this is a serious defect.

    Stalnaker argues that might implies epistemic possibility i.e. within an agent's power – 'John might have come to the party' i.e. within his power to. So Stalnaker thinks this possibility is outside the scope of the conditional i.e. 'It might be that if John had been invited, he would have come to the party'

    Remark: What about 'the coin might show up as heads' – there is no agent. Anyway, Stalnaker's solution is clearly contingent on a specific reading of counterfactual possibility i.e. epistemic possibility.